Weekly Project Update: Diffusion Diagnosis and Dataset Redesign

Date: May 14, 2026
Project: V2V BEV Reconstruction under Front-Region Sensor Occlusion

1. What I Led This Week

This week I moved the project from simply following model-level suggestions to making project-level decisions. The main shift was:

- I treated Diffusion as a scientific baseline instead of repeatedly forcing it to look good.
- I diagnosed why Diffusion failed using precision, recall, predicted occupancy rate, and empty false-positive rate.
- I redesigned the dataset protocol into a larger mixed-noise dataset with explicit `noisetype` and preprocessing metadata.
- I started mixed-noise U-Net and Pix2Pix training on Narval using the new `dataset_v2`.
- I built a paper-writing workflow so literature reading, notes, and thesis writing feed into each other.

The key conclusion is direct: **Diffusion is currently a negative baseline for this BEV reconstruction setup, while U-Net/Pix2Pix on the new mixed-noise dataset are now the next main experiments.**

2. Diffusion v3: Official Negative Baseline

Diffusion v3 was the first complete and reproducible Diffusion baseline. Compared with the earlier diffusion attempts, I fixed the training/evaluation protocol before judging it:

- Added timestep conditioning to the denoiser.
- Used real DDIM-style reverse sampling for validation/test instead of proxy evaluation.
- Added resume-safe checkpointing for Narval jobs.
- Used LR warmup + cosine decay and gradient clipping.
- Ran a full seed-42 Narval run for 120 epochs.

Diffusion v3 test result:

Metric	Value	Meaning
masked Occ-IoU	0.0467	Very low overlap with true occupied cells
masked precision	0.0467	Most predicted occupied cells are false positives
masked recall	1.0000	High only because it predicts almost everything occupied
masked F1	0.0893	Low balanced occupancy score
masked RMSE	0.3810	Poor reconstruction magnitude
fused full PSNR	17.00 dB	Full BEV looks better only because visible region is copied back

Interpretation: v3 is not just a little weak. It has a clear failure mode: high recall with very low precision means it predicts too much occupied space in the hidden region.

Diffusion v3 Visual Evidence

These examples use the correct inpainting/fused visualization: the visible ego region is preserved, and the Diffusion output is shown only in the hidden front sector. This is the version I should present to explain the failure fairly. The problem is not that the whole BEV image is replaced by noise; the problem is that the reconstructed hidden sector is noisy / over-occupied and does not recover clean multi-layer BEV structure.

3. Registered / Pre-Layer Diffusion Follow-Up

After v3 failed, I did not blindly rerun the same setup. I tested the teacher-inspired preprocessing idea as a controlled follow-up.

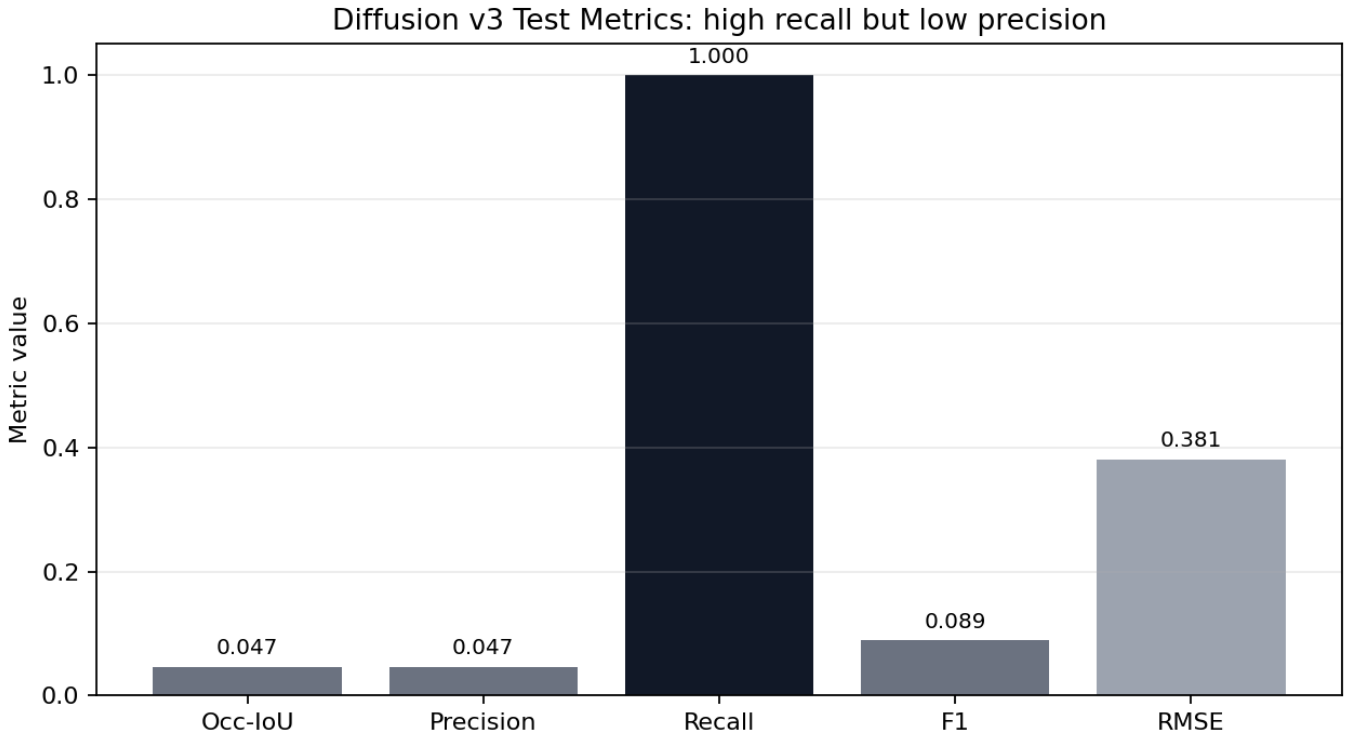


Figure 1: Diffusion v3 metric bars

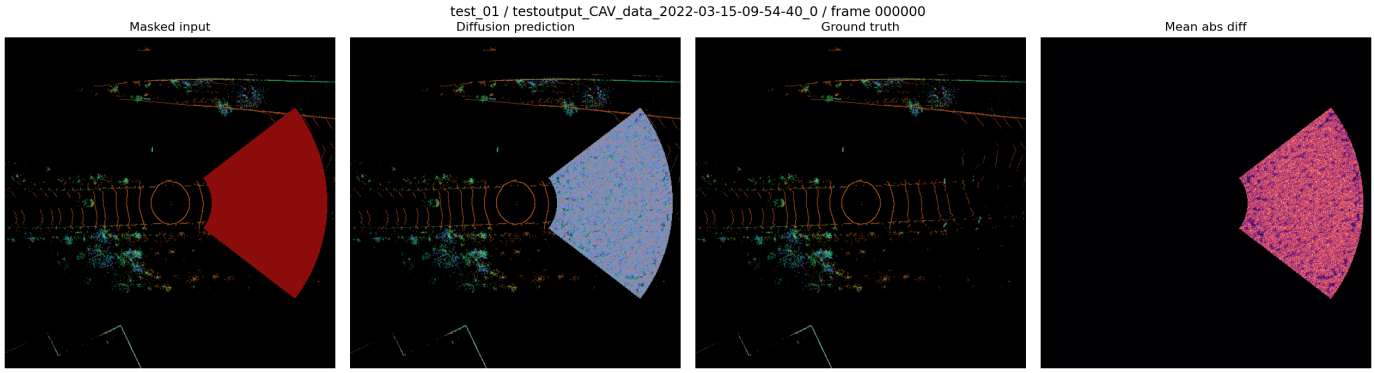


Figure 2: Diffusion v3 fused inpainting sample 1

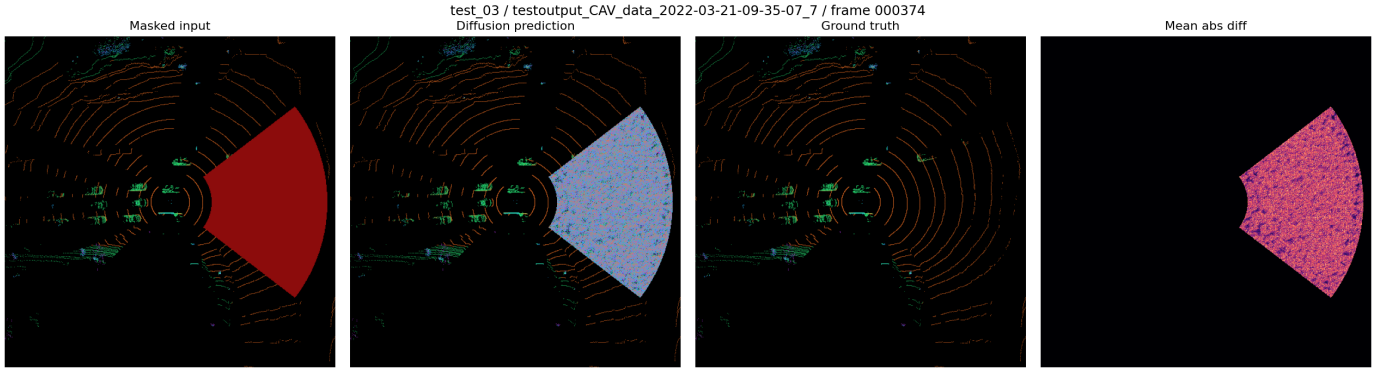


Figure 3: Diffusion v3 fused inpainting sample 2

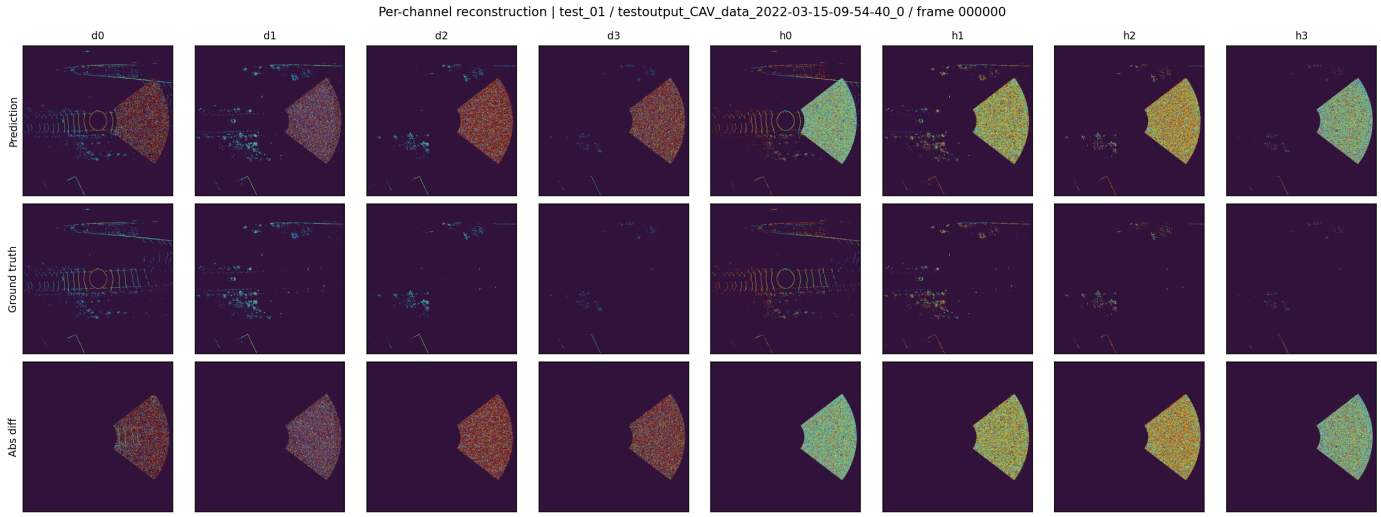


Figure 4: Diffusion v3 per-layer split

What Changed

I added a BEV-level preprocessing path:

- **Registration:** estimate a translation between neighbor BEV and visible ego BEV using occupancy centroid in the visible region.
- **Per-layer processing:** after alignment, calibrate each of the 8 BEV channels separately using visible-region statistics.
- **Why per-layer:** channels 0-3 are density-like occupancy bins, channels 4-7 are height-related bins. Treating them like RGB would be wrong.

I also changed the diffusion training objective to reduce the all-occupied failure:

Component	Setting	Reason
noise loss weight	0.2	Reduce pure denoising dominance
shared reconstruction weight	2.0	Force x0 reconstruction to matter more
empty penalty weight	1.0	Penalize predicting obstacles in truly empty hidden cells
height consistency	0.1	Preserve multi-layer height structure
occupancy BCE	0.0	Previous BCE/focal variants worsened all-occupied saturation

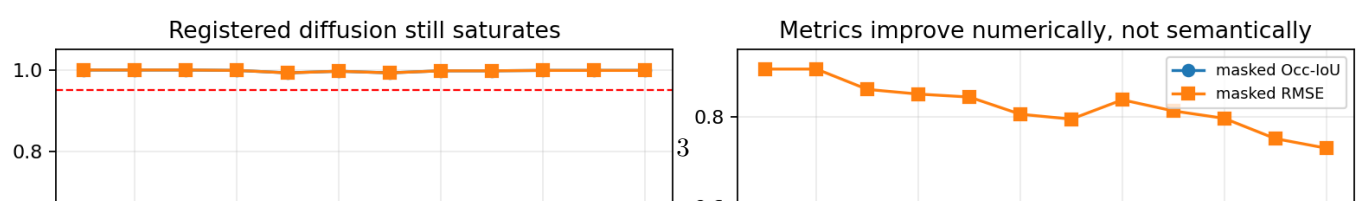
Run Evidence

Run root:

`/home/syin94/scratch/MEng_Project/runs/training_diffusion_registered_x0_seed42_v1`

Important log lines:

```
[regx0] Run root: /home/syin94/scratch/MEng_Project/runs/training_diffusion_registered_x0_seed42_v1
[regx0] train_diffusion sha256: a320cc647820c071a4abc67918511a1dde78aaafb0130c4f1b76b59de31d0f38
[regx0] dataset sha256: 32eb2ac259b3bb14446d8be56ca0af8f2d392b689cba69a285a9d41876483141
  epoch 5 done | val IoU=0.0447 prec=0.0447 rec=1.0000 val RMSE=0.906115 fusedPSNR=9.47 predOcc=1.000 emptyF
  epoch 30 done | val IoU=0.0447 prec=0.0447 rec=0.9971 val RMSE=0.806104 fusedPSNR=10.49 predOcc=0.997 empt
  epoch 60 done | val IoU=0.0447 prec=0.0447 rec=0.9994 val RMSE=0.730949 fusedPSNR=11.34 predOcc=0.999 empt
```



Registered v4 Visual Evidence

These panels show the teacher-inspired follow-up with per-layer neighbor registration and layer normalization. The visible ego region is still preserved, so the visual diagnosis is focused on the front hidden sector. Even after registration, the hidden sector remains over-filled/noisy instead of matching the sparse ground truth.

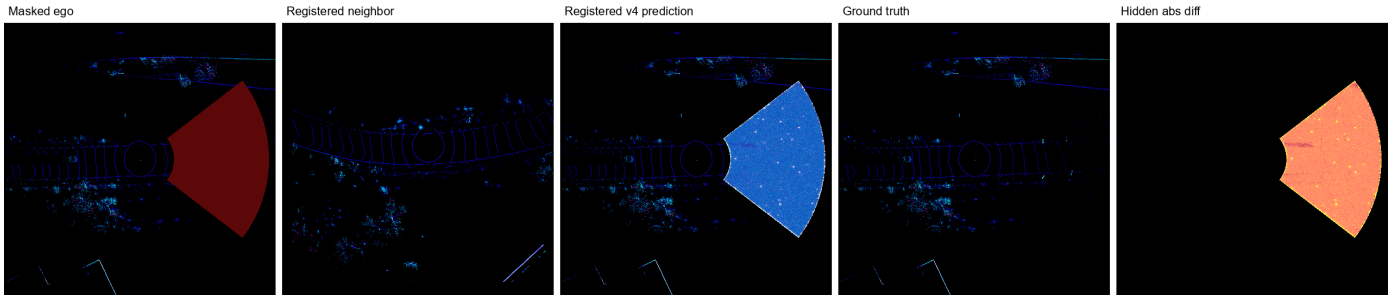


Figure 6: Registered v4 fused sample 1

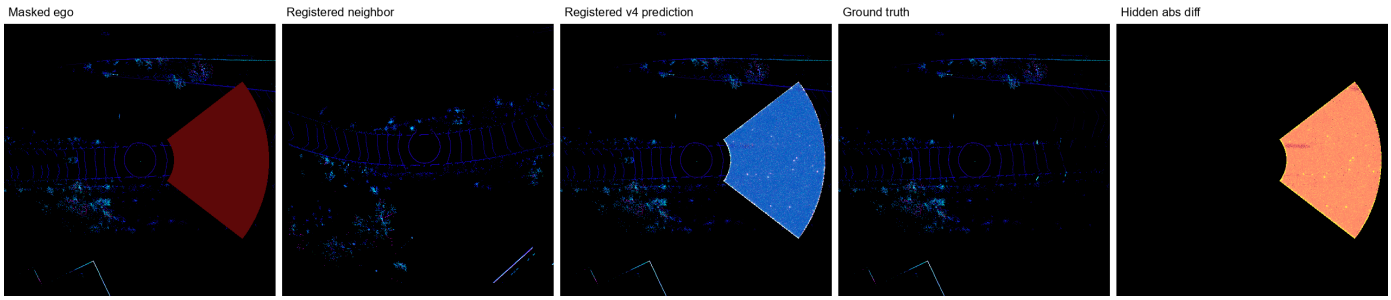


Figure 7: Registered v4 fused sample 2

Decision: I stopped the run around epoch 62 because the diagnostic metrics stayed saturated:

Epoch	Occ-IoU	Precision	Recall	RMSE	Pred Occ Rate	Empty FP Rate
5	0.0447	0.0447	1.0000	0.9061	1.000	1.000
15	0.0447	0.0447	1.0000	0.8611	1.000	1.000
30	0.0447	0.0447	0.9971	0.8061	0.997	0.997
45	0.0447	0.0447	0.9982	0.8134	0.998	0.998
60	0.0447	0.0447	0.9994	0.7309	0.999	0.999

Interpretation: registration and per-layer calibration improved the project depth, but they did not solve diffusion saturation. The model still predicted almost the entire hidden region as occupied. This supports reporting Diffusion as a negative baseline rather than spending more GPU time on blind reruns.

4. Dataset Redesign: From One Mask to a Mixed-Noise Dataset

I rebuilt the dataset protocol into `dataset_v2` without modifying the original `dataset_prepared` folder.

Why These Noise Types

Noise type	Why I chose it
`sector75`	Original front-sector sensor failure; keeps the official task definition.
`front_rect`	A regular front block occlusion; simple, controlled, and easy to explain.

Noise type	Why I chose it
`front_blob`	An irregular front occlusion; closer to non-uniform real-world occlusion while still deterministic.

I did not choose random whole-image masking or arbitrary scattered masks because the project is not generic image inpainting. The driving question is front-region BEV recovery when ego sensing is weak.

Dataset v2 Result

```
[dataset_v2] start=Thu 14 May 2026 12:42:32 AM EDT
  "base_samples": 9846,
  "manifest_rows": 59076,
  "materialize_noisy": true,
  "split_rows": {
[dataset_v2] size
4.0G    /home/syin94/scratch/MEng_Project/data/dataset_v2
  "base_samples": 9846,
  "manifest_rows": 59076,
  "materialize_noisy": true,
  "split_rows": {
```

Item	Value
base samples	9,846
noise types	3
preprocess types	2
manifest rows	59,076
train rows	42,630
val rows	4,488
test rows	11,958
storage used	about 4.0 GB

Important point: `dataset_v2` is a separate folder. The original dataset is unchanged.

5. Current Training Started on the New Dataset

I initially queued separate runs by mask type, but I corrected the protocol: the main run should train on one mixed dataset using `noise_type=all`.

Current submitted mixed-noise jobs:

```
=== SQUEUE ===
             JOBID                 NAME ST       TIME   TIME_LIMIT     NODELIST(REASON)
          60948542        unet_v2_all42 PD       0:00   1-00:00:00           (Priority)
          60948543        pix2_v2_all42 PD       0:00   1-00:00:00           (Priority)
=== SACCT ===
JobID|JobName|State|Elapsed|Timelimit|ExitCode
60948542|unet_v2_all42|PENDING|00:00:00|1-00:00:00|0:0
60948543|pix2_v2_all42|PENDING|00:00:00|1-00:00:00|0:0
=== LOG TAIL U ===
=== LOG TAIL P ===
=== RUN DIRS ===
```

Main run configuration:

Model	Dataset	Noise setting	Preprocess	Seed	Epochs
U-Net	dataset_v2	all noise types mixed	none	42	80

Model	Dataset	Noise setting	Preprocess	Seed	Epochs
Pix2Pix	dataset_v2	all noise types mixed	none	42	80

This is the right next step because it tests whether the model can learn a general recovery policy over multiple front-region occlusion shapes.

6. Paper Writing Workflow

I also built a local paper/thesis workflow so reading and writing are not separate from experiments.

Local workspace:

`D:/MEng_Project/thesis`

The workflow is:

1. Read one paper.
2. Write one note in `thesis/notes/papers/`.
3. Track it in `references/reference_matrix.csv`.
4. Synthesize notes into LaTeX chapters.
5. Run citation and build checks.

This helps me lead the project because each experiment can be tied to literature, assumptions, and paper-writing sections instead of staying as isolated training runs.

7. Next Steps

1. Monitor the mixed-noise U-Net and Pix2Pix runs.
2. Evaluate the trained mixed models both overall and split by `sector75`, `front_rect`, and `front_blob`.
3. Compare mixed-training performance against the old single-mask baseline.
4. Run `register_layernorm` as a fair preprocessing ablation for U-Net and Pix2Pix, not only Diffusion.
5. Keep Diffusion v3 as the official negative baseline unless a later architecture-level method is justified.

8. Supervisor-Level Takeaway

My takeaway this week is that the main contribution is no longer just testing three model families. The project is becoming a structured BEV reconstruction study with:

- a confirmed strong baseline family: U-Net / Pix2Pix,
- a diagnosed negative generative baseline: Diffusion,
- a richer mixed-noise dataset protocol,
- explicit preprocessing metadata,
- and a paper workflow connecting experiments to writing.

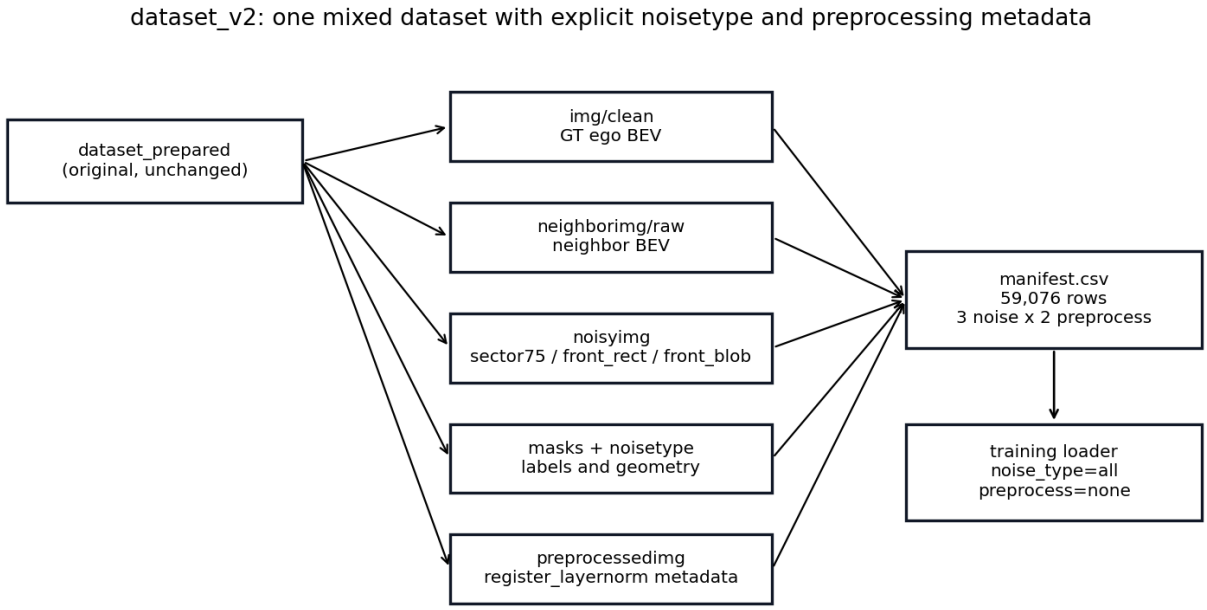


Figure 8: dataset_v2 layout

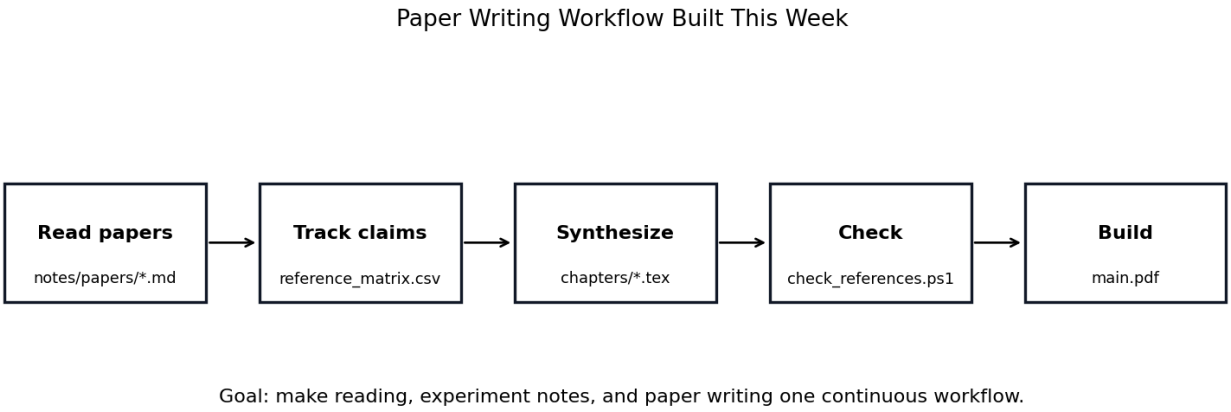


Figure 9: Paper workflow